

Services Consultant Architect Practice Lab 1

IMPORTANT: The **approximate build time for this lab is 5 minutes**. Please be patient. If the lab does not build after 15 minutes, try refreshing the page and rebuilding it again. If it does not build on the second attempt, contact ellearn@splunk.com.

Lab Overview

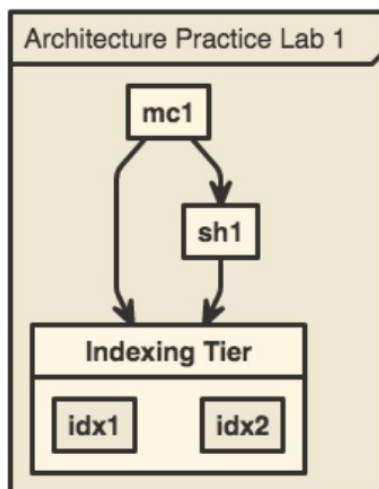
Maximum Estimated Duration = 3 Hours

This lab will focus on the download and installation of Splunk on 4 hosts.

You will configure a single Search Head, two Indexers, and a Monitoring Console.

The Indexers will be configured as distributed search peers of the Search Head and the Monitoring Console.

Host Name	Role
Ap1-xxxx-sh1	Search Head
Ap1-xxxx-mc1	Monitoring Console
Ap1-xxxx-idx1	Indexer
Ap1-xxxx-idx2	Indexer



NOTE: Do not reboot the instance OS during the lab. Doing so will cause the firewall to enable, thus blocking access to Splunk Web. If you cannot access Splunk Web on an instance, ssh into the instance and disable the firewall using the following command: `sudo systemctl stop ufw`.

Install Splunk on Your First Host (Search Head)

1. Access the Command Line Interface on your host.
2. Connect via ssh to your first host.
3. Connect using sccStudent user using the password provided in your instance list.
ssh sccStudent@yourhostname.students.splunk.education

Create a splunk user

1. Create a splunk user and password on your host server sudo
sudo adduser --disabled-password splunk (accept all defaults)

Unarchive Splunk

NOTE: The Splunk Enterprise installation package has been staged in the /opt folder on the lab systems.

1. Change to the /opt directory and issue the following command.
sudo tar -xvzf splunk_package_name.tgz

NOTE: Further information on the .tgz installation can be found here:
<http://docs.splunk.com/Documentation/Splunk/latest/Installation/InstallonLinux>

Install Splunk as root

1. As the root user start Splunk for the first time.
Cd /opt/splunk/bin
./splunk start --accept-license

When prompted for the password for the default admin account, **use the password provided in your Server list.**

2. Once Splunk has started you will need to launch Splunk Web.
3. After launching Splunk Web issue the following search and confirm events are present:
a index=_internal

Run Splunk as a Non-Root User

In most enterprise environments, for security reasons, you will not want to run Splunk as the root user. Let's change the ownership to the splunk user.

The splunk user and group have already been configured on all hosts.

1. To run Splunk as a non-root user follow these directions:
<https://docs.splunk.com/Documentation/Splunk/8.2.6/Installation/RunSplunkasadifferentor-non-rootuser>
2. Log back into Splunk Web and run the test search again.

Enable Splunk to Start on Boot as the splunk user

Finally, we'll want to enable splunk to restart automatically as the non-root user splunk. To enable this you will require root privileges.

1. To enable boot-start follow these directions:
<https://docs.splunk.com/Documentation/Splunk/8.2.6/Admin/ConfigureSplunktostartatboottime>
- Install Splunk on your remaining instances (idx1, idx2, mc1)

Using the same instructions as above install Splunk on the remaining instances:

1. Access the command line interface on your host.
2. Create your splunk user.
3. Untar Splunk Enterprise
4. Install Splunk as root.
5. For the admin password, **use the password found in your Server list.**
6. Run Splunk as splunk user.
7. Enable Splunk to start on boot as the splunk user.

Configure Your Indexers

First, let's understand what an index is.

1. Read the documentation on the following links:

<http://docs.splunk.com/Documentation/Splunk/latest/Indexer/Aboutindexesandindexers>
<http://docs.splunk.com/Documentation/Splunk/latest/Indexer/Howindexingworks>

Your indexer will need to be able to receive data from other forwarders.

2. Read the documentation on the following link.

<http://docs.splunk.com/Documentation/Splunk/latest/Forwarding/Enableareceiver>

Instead of using any of the methods in the document above, you will need to use the appropriate base configuration app to configure your indexer's input configuration.

The base configuration app templates can be found here:

<https://drive.google.com/drive/folders/107qWrfsv17j5bLxc21ymTagjtHG0AobF>

To configure your indexer:

3. Copy the `org_all_indexer_base` template app to your `/opt/splunk/etc/apps` folder.

This template has an `inputs.conf` with a listening port preconfigured.

4. Rename the app to `ap1_all_indexer_base`.

This sort of rename should **always be performed** on any base template apps implemented.

Always check the contents of app templates before applying them.

5. Go through each configuration file and validate the settings that are enabled and uncommented.
6. Ensure the permissions on your copied apps are owned by the `splunk` user.
7. Restart Splunk to apply the configurations from the template app.
8. Check the receiver status with the following command as the `splunk` user:
`/opt/splunk/bin/splunk display listen`
9. Repeat the same steps on your second indexer.

Configure Distributed Search

For your Search Head to be able to search the data stored on your indexers, you will need to configure Distributed Search.

Information on Distributed Search can be found here:

<http://docs.splunk.com/Documentation/Splunk/latest/DMC/DMCoverview>

1. To configure Distribute Search, follow the instructions here using either the CLI or Splunk Web:
<http://docs.splunk.com/Documentation/Splunk/latest/DistSearch/Configureddistributedsearch>
2. Repeat the same process to add your indexers to your Monitoring Console instances as a search peer.
3. Ensure that you also add the search head as a distributed search peer to the MC.

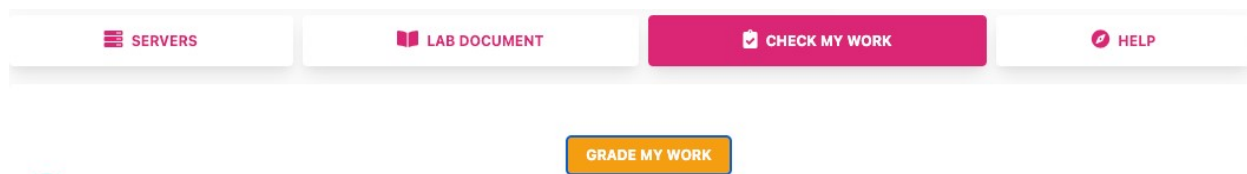
Configure Monitoring Console

To monitor your configured environment, you will need to configure your monitoring console in distributed mode.

1. Click the link below to learn more about the Monitoring Console:
<http://docs.splunk.com/Documentation/Splunk/latest/DMC/DMCoverview>
2. Read the instructions on configuring the monitoring Console in distributed mode; click the link below:
<http://docs.splunk.com/Documentation/Splunk/latest/DMC/Configureindistributedmode>

Grading Your Work

Grading of the lab is done through the same interface where your student instances are displayed.



Grading of the labs is an intensive process involving many calls to the instances. This may take several minutes to complete. Please be patient. If the grading does not complete in 5 minutes, please reach out to ellearn@splunk.com for assistance.

For best results, make sure you complete all of the lab steps before grading the lab.

1. Click the Check My Work Tab.
2. Click the Grade My Work button.



After a few minutes, your results will be displayed.

You must pass all graded items to receive credit for the lab. Resolve any failed grading items and re-issue the grading check.